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Samuel J. Armstrong

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EDUCATION

PhD in Mathematics, University of Kansas
Lawrence, KS

2026-Present

MA in Mathematics, University of Kansas
Lawrence, KS

2026

BS in Mathematics, Buena Vista University
Storm Lake, IA

2024

RESEARCH INTERESTS

Partial Differential Equations, Numerical Analysis

I am broadly interested in the analysis and numerics of partial differential equations that arise from fluid mechanics. Currently, my research investigates the regularity and stability of solutions of a parameterized family of nonlinear wave equations. Such a family includes some well known equations such as the Hunter-Saxton, Camassa-Holm, and Novikov equations.

PUBLICATIONS

1. S. J. Armstrong, G. Chen, T. Huang, and Y. Shen. *Regularity of Structurally Stable Cusp Singularities for Two Systems of Quasilinear Wave-type Equations*. Submitted.

TEACHING EXPERIENCE

Fall 2026 Lab Instructor: MATH 127, Calculus III

Spring 2026 Teaching Assistant: MATH 146, Calculus II (Honors)

Spring 2026 Instructor: MATH 126, Calculus II (Enhanced)

Fall 2025 Instructor: MATH 115, Calculus I (Business)

Summer 2025 Instructor (Online): MATH 104, Precalculus

Spring 2025 Instructor: MATH 115, Calculus I (Business)

Fall 2024 Teaching Assistant: MATH 101, College Algebra

SERVICE

5. *Math Graduate Student Organization President*. University of Kansas, 2026-2028.
 4. *Analysis Study Group Leader*. University of Kansas, Summer 2026.
 3. *MSAM Mathematics and Statistics Competition Volunteer*. University of Kansas, 2026.
 2. *First-year Graduate Student Mentor*. University of Kansas, 2025-2027.
 1. *Math Graduate Student Organization Member-at-Large*. University of Kansas, 2025-2026.
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PRESENTATIONS

5. *Generic Regularity and Singular Behavior for a Generalized Hunter-Saxton Type Equation*. Workshop for Research and Workforce Development in Fluid Mechanics II, Lincoln, NE, 2026.
 4. *Structurally Stable Singularities for a Generalized Hunter-Saxton Type Equation*. University of Kansas Differential Equations, Dynamical Systems & Geometric Analysis Seminar, Lawrence, KS, 2026.
 3. *An Introduction to Physics-Informed Neural Networks*. University of Kansas Math Graduate Student Organization Seminar, Lawrence, KS, 2025.
 2. *Swimming with Deep Learning*. Biofluids: Low Re Swimming II, APS 76th Annual Meeting of the Division of Fluid Dynamics, Washington, DC, 2023.
 1. *Grid Codes for Secure Distributed Matrix Multiplication*. AMS Contributed Paper Session on Computer Science, Joint Mathematics Meetings, Boston, MA, 2023.
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AWARDS

2. *U.G. Mitchell Graduate Summer Scholarship*. University of Kansas, 2026.
1. *Mathematics Senior of the Year*. Buena Vista University, 2024.